Notes: Di Maggio et al. (AER, 2017)

Interest Rate Pass-Through: Mortgage Rates, Household Consumption, and Voluntary Deleveraging

Contents

1 Big picture

- **Question.** How do lower mortgage rates pass through to household consumption, voluntary debt repayment, mortgage default, and local real activity?
- **Setting.** The paper studies U.S. borrowers with hybrid adjustable-rate mortgages (ARMs) originated in 2005–2007. These mortgages have an initial fixed-rate period and then automatically reset to market-indexed rates.
- **Core empirical idea.** When short-term interest-rate indices collapsed after the financial crisis, five-year ARMs originated in 2005–2007 reset during 2010–2012. The resets produced large, automatic reductions in monthly mortgage payments.
- **Main identification advantage.** The decline in payments is generated by pre-existing mortgage-contract rules, not by a household's refinancing decision. This avoids the usual endogeneity of refinancing, creditworthiness, housing equity, and local economic conditions.
- **Main data contribution.** The authors merge monthly mortgage-servicing data with monthly credit-bureau data. This lets them observe mortgage payments, mortgage balances, FICO scores, auto-loan balances, and other debt at the household-month level.
- **Main treatment magnitude.** In the non-agency five-year ARM sample, average monthly mortgage payments fall from about \$1,921 to about \$916 after reset, a decline of roughly \$940 or about 53%.
- **Main consumption result.** Borrowers increase new car spending by about \$77 per month in the first year after reset and \$108 per month in the second year after reset. The probability of buying a car rises by about 25–35% relative to the pre-reset monthly purchase probability.
- **Main deleveraging result.** Borrowers also use part of the liquidity shock to repay mortgage principal faster. Voluntary mortgage repayment rises by about \$68 per month in the first year and \$75 per month in the second year.
- **Main heterogeneity result.** Lower-income and lower-housing-wealth borrowers consume more out of the payment reduction and deleverage less. This is consistent with life-cycle models in which households with lower wealth or income have higher marginal propensities to consume.
- **External validity result.** Similar patterns appear for conforming agency ARMs, although the payment reduction is smaller: about \$280 per month, or 23% of the initial payment.
- **Regional result.** Zip codes with higher pre-existing ARM shares experience larger mortgage-rate declines and subsequently show relative declines in delinquency, stronger house-price growth, stronger auto-sales growth, and higher employment growth.
- **Economic takeaway.** Monetary-policy pass-through depends on household balance sheets and mortgage-contract rigidity. ARMs transmit lower interest rates automatically, while fixed-rate mortgages require refinancing and therefore can block pass-through for constrained or underwater borrowers.

2 Data and measurement (mortgage contracts and household outcomes)

2.1 Why adjustable-rate mortgages are the right laboratory

- Hybrid ARMs have an initial fixed rate followed by automatic resets based on a reference index such as LIBOR or a Treasury rate.
- Five-year ARMs originated in 2005, 2006, and 2007 first reset in 2010, 2011, and 2012.
- The reset happened after short-term interest rates had fallen sharply, so the reset lowered interest rates and scheduled payments for many borrowers.
- The reduction was automatic. Borrowers did not need to refinance, pay closing costs, qualify for a new loan, or have positive home equity.
- This setting turns the ARM reset into a quasi-experimental household liquidity shock.

Interpretation. The key treatment is not “a borrower chose a lower rate.” It is “a contract written years earlier mechanically changed the payment when the reset date arrived.” This distinction is what makes the design powerful.

2.2 Borrower-level mortgage and credit-bureau data

- The main mortgage data come from BlackBox Logic, which covers a large share of privately securitized non-agency mortgages from this period.
- The authors focus on prime borrowers, owner-occupied residences, and loans originated during 2005–2007.
- The mortgage data report loan type, origination balance, initial rate, initial FICO score, monthly payment, current balance, delinquency status, and location.
- Mortgage records are merged with Equifax credit-bureau reports.
- Equifax provides monthly information on auto loans, mortgage debt, home-equity debt, monthly payment histories, updated FICO scores, and other household liabilities.

Interpretation. The paper does not merely observe whether a mortgage rate fell. It observes how the same household’s balance sheet evolves month by month before and after the reset.

2.3 Main sample: non-agency five-year ARMs

- The main sample contains **46,578** borrowers with non-agency five-year ARMs.
- These loans are interest-only for the first ten years. This makes voluntary principal repayment easier to identify because scheduled principal amortization is limited.

- Mean origination FICO is **723.3**.
- Mean loan balance is about **$357,949**.
- Mean initial loan-to-value ratio is about **77.1%**.
- Mean initial mortgage rate is **6.44%**; mean rate after adjustment is **3.09%**.
- Mean monthly payment is **$1,921** before reset and **$916** after reset.

Interpretation. The reset creates a very large income shock for treated households: roughly one thousand dollars per month for a typical borrower in the main sample.

2.4 Control-group and external-validity mortgage samples

- The paper uses **26,543** non-agency ten-year ARM borrowers as a control group in the difference-in-differences design.
- Ten-year ARMs were originated during the same broad period but do not reset at year five.
- For external validity, the authors also use conforming agency ARM data from a large secondary-market participant.
- The conforming sample contains **71,741** five-year ARM borrowers and **21,466** seven-year ARM borrowers.
- Conforming five-year ARMs have average loan balances of about **$201,422**, average initial rates of **5.78%**, and average post-reset rates of **3.28%**.
- Their monthly payments fall from about **$1,175** to **$902**, a reduction of about **23%**.

Interpretation. The non-agency sample gives a strong treatment shock; the conforming sample shows that the same mechanism also operates among more typical agency mortgage borrowers.

2.5 Consumption measure: new car purchases

- The main consumption proxy is new car spending financed by auto debt.
- The authors infer car purchases from discrete increases in auto-loan balances.
- The dollar amount of new car spending is the change in auto-loan balance at the time of purchase.
- In the non-agency five-year ARM sample, average monthly new car spending is about **$305**.
- The monthly probability of buying a new car is about **1.4%**.
- The authors note that this measure understates total consumption because it misses cash purchases, checking-account spending, and other non-credit forms of spending.

Interpretation. Car purchases are a high-frequency durable-spending measure. They are especially useful because auto-loan balances in credit-bureau data reveal purchase timing.

2.6 Voluntary deleveraging measure

- The paper measures voluntary mortgage debt repayment using changes in mortgage balances.
- Because the main sample consists of interest-only loans during the relevant period, reductions in principal are interpreted as voluntary repayments rather than scheduled amortization.
- Full payoffs are excluded because they often reflect refinancing or house sales rather than voluntary deleveraging.
- In the non-agency five-year ARM sample, average voluntary mortgage repayment before reset is about **$52** per month.

Interpretation. Deleveraging is the part of the liquidity shock that households use to repair balance sheets instead of consuming immediately.

2.7 Regional data and ARM exposure

- The regional analysis combines BlackBox data with LPS/Black Knight mortgage data to measure local mortgage characteristics.
- The key local exposure variable is the zip-code share of outstanding mortgages that are ARMs, measured before the large rate declines.
- The paper also uses Equifax Credit Trends, American Community Survey demographics, Zillow house-price indices, R. L. Polk auto-sales data, and Census ZIP Business Patterns employment data.
- The matched regional sample contains **1,000** zip codes, split between high- and low-ARM-share zip codes after propensity-score matching.
- High-exposure zip codes have mean ARM shares of **35.2%**; low-exposure zip codes have mean ARM shares of **17.3%**.

Interpretation. Borrower-level evidence identifies the direct household response. Regional evidence asks whether areas more exposed to automatic rate reductions experience broader local improvements.

3 Models and empirical specifications (all equations plus interpretation)

3.1 Main household event-study specification

The main household-level specification is

$$Y_{i,t} = \alpha_i + \sum_{\theta=-1}^{2} \beta_\theta \mathbf{1}_{i,t\in\theta} + \Gamma X_{i,t} + \varepsilon_{i,t}, \tag{1}$$

where i indexes households, t indexes months, and θ denotes the time period relative to the first interest-rate adjustment.

- $Y_{i,t}$ is a monthly household outcome: scheduled mortgage payment, new car spending, car-purchase indicator, voluntary mortgage repayment, or default.
- α_i is a borrower fixed effect.
- $\mathbf{1}_{i,t\in\theta}$ indicates whether a month lies in a particular window around reset.
- The paper reports coefficients for one year before reset, one year after reset, and two years after reset.
- $X_{i,t}$ includes borrower, mortgage, and local controls, including updated FICO, current LTV, zip-code house-price controls, county-month fixed effects, and cohort-time fixed effects in relevant specifications.
- Standard errors are clustered at both the borrower and month level.

Interpretation. The regression compares the same borrower before and after an automatic payment reset, while netting out common local-month shocks and borrower fixed differences.

3.2 Dynamic quarterly event-study version

The paper also estimates a quarterly event-study version:

$$Y_{i,t} = \alpha_i + \sum_{q\neq -1} \beta_q \mathbf{1}_{i,t\in q} + \Gamma X_{i,t} + \varepsilon_{i,t}, \tag{2}$$

where q indexes quarters relative to the reset, and the pre-reset quarter is the omitted category.

Interpretation. The quarterly event study checks whether outcomes move before the reset. A clean design should show little pre-trend in payments, car spending, or debt repayment before the mechanical reset, followed by sharp changes afterward.

3.3 Heterogeneity specification

To study different responses across household types, the paper interacts the event-time indicators with a borrower-type dummy X_i:

$$Y_{i,t} = \alpha_i + \sum_{\theta} \beta_\theta \mathbf{1}_{i,t\in\theta} + \sum_{\theta} \delta_\theta \left(\mathbf{1}_{i,t\in\theta} \times X_i\right) + \Gamma X_{i,t} + \varepsilon_{i,t}. \tag{3}$$

The paper uses indicators for high income, high LTV, and high FICO.

Interpretation. The coefficient β_θ gives the response of the baseline group. The interaction coefficient δ_θ says whether high-income, high-LTV, or high-FICO borrowers respond differently after the same kind of mortgage-payment shock.

3.4 Difference-in-differences specification with longer-reset ARMs

To address mortgage-age trends, the paper compares five-year ARMs with longer-reset ARMs originated in the same broad period:

$$\begin{aligned} Y_{i,t} = \alpha_i + \gamma_{5Y}\mathbf{1}_{i\in 5Y} + \sum_{\theta=-4}^{5}\left[\gamma_\theta \mathbf{1}_{i,t\in\theta} + \delta_\theta\left(\mathbf{1}_{i\in 5Y}\times \mathbf{1}_{i,t\in\theta}\right)\right] \\ + \Gamma X_{i,t} + \varepsilon_{i,t}. \end{aligned} \tag{4}$$

- The treatment group is five-year ARM borrowers.
- The control group is ten-year ARMs in the non-agency sample and seven-year ARMs in the conforming sample.
- The event date is month 60 after origination, when five-year ARMs reset.
- The key coefficients are δ_θ, the differential change for five-year ARMs relative to longer-reset ARMs.

Interpretation. This design asks whether five-year ARM borrowers change behavior at month 60 relative to otherwise similar borrowers whose ARMs have not yet reset.

3.5 Regional reduced-form specification

The regional analysis uses a cross-zip-code specification of the form

$$\Delta y_z = a + b\,\mathrm{ARMShare}_{z,2006} + \Pi Z_z + \lambda_s + u_z, \tag{5}$$

where Δy_z is the change in a regional outcome between the pre-decline and post-decline periods, $\mathrm{ARMShare}_{z,2006}$ is the pre-existing ARM share in zip code z, Z_z is a vector of zip-code controls, and λ_s is a state fixed effect.

Interpretation. Zip codes with more ARMs should experience larger effective mortgage-rate cuts when market rates fall. The coefficient b measures whether those areas also have differential changes in delinquency, house prices, auto sales, and employment.

3.6 Regional instrumental-variable strategy

The paper also instruments local ARM share using the historical share of home transactions with prices below 1.25 times the conforming loan limit:

$$\mathrm{ARMShare}_z = \pi_0 + \pi_1 \mathrm{ConfEligibleShare}_{z,1998-2002} + \Pi Z_z + \lambda_s + v_z, \tag{6}$$

$$\Delta y_z = a + b\,\widehat{\mathrm{ARMShare}}_z + \Pi Z_z + \lambda_s + u_z. \tag{7}$$

Interpretation. Areas where homes were more often conforming-loan eligible should have had lower ARM shares because ARMs were less popular among GSE-backed loans. The exclusion restriction is that this historical conforming-eligibility share affects 2009–2012 outcomes mainly through ARM exposure, not through another direct channel.

3.7 Imputing total consumption from auto spending

The paper uses a back-of-the-envelope scaling exercise:

$$\Delta \text{TotalConsumption} = \frac{\text{TotalConsumption}}{\text{CarSales}} \times \frac{\beta_{\text{TotalConsumption}}}{\beta_{\text{CarSales}}} \times \Delta \text{CarSales}. \tag{8}$$

Using the numbers in the paper,

$$\Delta \text{TotalConsumption} = \frac{1}{0.045} \times \frac{0.7}{2.3} \times \$110 \approx \$744. \tag{9}$$

Interpretation. If the auto-spending response scales to total spending like it does under state-level shocks, a \$940 monthly payment reduction implies roughly \$744 in extra total consumption, or an overall marginal propensity to consume of about 0.8.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the borrower-level estimates

- The timing of ARM resets is predetermined by the mortgage contract.
- Interest-rate indices unexpectedly fell to very low levels during the reset period.
- Borrowers with the same contract type reset at different calendar dates because they originated at different dates.
- The design compares borrowers who have already reset to similar borrowers who have not yet reset.
- Borrower fixed effects absorb time-invariant differences in preferences, risk, and balance-sheet levels.

Interpretation. The clean source of variation is not cross-sectional ARM ownership versus fixed-rate mortgage ownership. It is within-ARM timing around a mechanical reset.

4.2 Why refinancing endogeneity is avoided

- Standard refinancing is endogenous because borrowers choose whether to refinance based on credit scores, home equity, income, expectations, and closing costs.
- During the crisis, many borrowers were underwater or credit constrained, so refinancing access was limited.

- ARM resets occur without borrower action and without a new underwriting decision.
- This makes the payment cut available even to borrowers who might not qualify for refinancing.

Interpretation. The paper isolates the effect of a payment reduction itself, rather than the effect of selecting into refinancing.

4.3 Role of fixed effects and controls

- Borrower fixed effects control for stable household traits.
- County-month fixed effects absorb local macroeconomic shocks, local labor-market conditions, and housing-market changes that vary over time.
- Cohort-time fixed effects allow loans originated in different years to have different time patterns.
- Updated FICO and current LTV controls absorb time-varying borrower credit and balance-sheet conditions.

Interpretation. A central concern is that borrowers resetting in 2010 may live in different local economies than borrowers resetting in 2012. County-month and cohort-time controls are designed to remove those confounds.

4.4 Evidence against pre-trends

- Event-study figures show mortgage payments are flat before reset and drop sharply at reset.
- Voluntary repayment does not show a meaningful pre-trend before reset.
- Car spending rises slightly just before reset, plausibly because borrowers receive advance notice of the lower payment, but the main response occurs after reset.
- Difference-in-differences results also show much larger changes only when five-year ARMs reset relative to longer-reset ARMs.

Interpretation. The empirical pattern is difficult to explain with smooth mortgage-age trends alone. The sharp timing around reset supports a causal interpretation.

4.5 Why the difference-in-differences design matters

- The main event study cannot perfectly separate reset timing from mortgage age.
- Longer-reset ARMs provide a control group that is similar in origination period and mortgage type but does not experience a year-five reset.
- The DiD specification shows that five-year ARMs have differential declines in payments and differential increases in car spending and mortgage repayment at the reset date.

Interpretation. If all mortgages naturally changed consumption around month 60, the DiD design would not show a sharp five-year ARM effect relative to longer-reset ARMs.

4.6 Attrition and sample-selection concerns

- Some borrowers leave the sample because of payoff, refinancing, liquidation, foreclosure, or sale.
- The paper discusses attrition and studies active loans separately from paid-off and liquidated loans.
- Paid-off loans tend to have lower current LTVs, while liquidated loans are associated with worse housing-equity positions.
- The authors emphasize that the main estimates describe borrowers who remain active around the reset window.

Interpretation. Attrition is a limitation, but the mechanism is clear: payment reductions are observed for active borrowers. If distressed borrowers leave before reset, the estimated consumption effects may understate the potential effect for highly constrained households who would have benefited from the reset.

4.7 External validity

- Non-agency ARMs have large balances and large treatment shocks.
- Conforming agency ARMs are more representative of typical U.S. mortgage borrowers.
- The conforming sample shows qualitatively similar responses despite smaller payment reductions.
- Regional evidence uses local ARM shares to move beyond borrower-level credit records and capture broader local outcomes.

Interpretation. The paper is strongest when the same mechanism appears in three places: non-agency borrower-level data, conforming borrower-level data, and regional aggregate outcomes.

4.8 What the paper can and cannot distinguish

- The paper credibly estimates responses to a persistent reduction in required debt service.
- It does not observe all consumption categories directly; the main direct measure is debt-financed car purchases.
- It does not fully estimate economy-wide general-equilibrium effects because regional regressions compare high- and low-exposure areas.
- It shows that contract structure affects pass-through, but it does not claim that ARMs are always optimal for all households in all states of the world.

Interpretation. The paper is best read as evidence on the household-balance-sheet channel of monetary policy and on the importance of mortgage-contract pass-through.

5 Tables and figures

5.1 Table 1: Summary statistics for non-agency and conforming ARMs

Read:

- Non-agency five-year ARMs: 46,578 borrowers, average FICO 723, loan balance $357,949, LTV 77.1%, and initial rate 6.44%.
- After reset, non-agency five-year ARM rates fall to 3.09%, and monthly payments fall from $1,921 to $916.
- Non-agency ten-year ARMs are larger, with average balances of $536,342 and average monthly payments of $2,700.
- Conforming five-year ARMs: 71,741 borrowers, average FICO 711.5, loan balance $201,422, and LTV 76.0%.
- After reset, conforming five-year ARM payments fall from $1,175 to $902.

Economic message. The reset is large in the non-agency sample and meaningful in the conforming sample. The paper has both a high-intensity treatment sample and a more representative external-validity sample.

TABLE 1—SUMMARY STATISTICS FOR BORROWERS WITH NON-AGENCY AND CONFORMING ARMS

	Borrowers with five-year ARMs		Borrowers with ten-year ARMs	
	Mean	SD	Mean	SD
Panel A. Summary statistics for borrowers with non-agency ARMs				
FICO	723.3	39.4	736	39.7
Loan balance	357,949	271,600	536,342	347,622
Loan-to-value (LTV) ratio	77.11	10.01	72.82	12.05
Initial interest rate	6.44	0.76	6.14	0.52
Average monthly payment	1,921	1,471	2,700	1,819
Monthly new car spending	305.1	3,161	364.4	3,711
Fraction of borrowers buying new car per month	0.014	0.116	0.015	0.121
Voluntary mortgage debt repayment per month	52.21	400.1	88.49	619.6
Interest rate after adjustment for five-year ARMs	3.09	0.48	—	—
Monthly payment after adjustment for five-year ARMs	915.8	721.9	—	—
Number of borrowers	46,578		26,543	
	Borrowers with five-year ARMs		Borrowers with seven-year ARMs	
Panel B. Summary statistics for borrowers with conforming ARMs				
FICO	711.5	57.57	712.03	61.12
Loan balance	201,422	86,470	200,044	87,590
Loan-to-value (LTV) ratio	76.00	14.59	74.98	15.97
Initial interest rate	5.78	0.79	5.73	0.68
Average monthly payment	1,175	507	1,163	512
Monthly new car spending	166.2	2,310	172.32	2,252
Fraction of borrowers buying new car per month	0.008	0.092	0.009	0.094
Voluntary mortgage debt repayment per month	27.18	157.7	30.77	167.0
Interest rate after adjustment for five-year ARMs	3.28	0.45	—	—
Monthly payment after adjustment for five-year ARMs	902.3	408.1	—	—
Number of borrowers	71,741		21,466	

Notes: This table reports descriptive statistics for the main variables employed in the analysis. Panel A presents the main characteristics of a sample of borrowers with non-agency adjustable-rate mortgages (ARMs) having the first interest rate adjustment after five and after ten years from origination date, respectively. Panel B presents similar characteristics for a sample of borrowers with conforming ARMs having the first interest rate adjustment after five and after seven years from origination date, respectively.

Source: BlackBox Logic for borrowers with non-agency ARMs originated in 2005–2007 and a large secondary market participant for borrowers with conforming ARMs originated during the period 2005–2007.

TABLE 2—IMPACT OF RATE REDUCTIONS ON MONTHLY MORTGAGE PAYMENTS, NEW CAR SPENDING, AND VOLUNTARY DEBT REPAYMENT

	Mortgage payment (1)	Mortgage payment/initial payment (2)	New car spending (3)	New car spending/initial payment (4)	Probability of new car purchase (5)	Voluntary debt repayment (6)	Debt Repayment/ initial payment (7)
One year before	−11.50	−0.0036	23.36	0.006	0.144	1.92	0.000
	(5.65)	(0.0008)	(10.01)	(0.0075)	(0.0339)	(1.74)	(0.001)
One year after	−943.4	−0.528	77.09	0.043	0.345	68.25	0.041
	(34.79)	(0.0058)	(12.92)	(0.0106)	(0.0440)	(3.776)	(0.0023)
Two years after	−937.6	−0.529	108.2	0.072	0.478	74.92	0.044
	(27.12)	(0.0065)	(18.92)	(0.0165)	(0.0851)	(6.307)	(0.004)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,222,930	2,222,930	2,223,476	2,223,476	2,223,476	2,037,981	2,037,981
R^2	0.704	0.737	0.001	0.001	0.001	0.006	0.004

Notes: This table presents the estimates from the OLS regressions that examine the relationship between the timing relative to the first interest rate reset occurring five years after the loan's origination and a range of outcomes in a sample of borrowers with non-agency five-year ARMs. The main independent variables are dummies identifying different time periods before and after the reset date. One year before identifies the 12 months before up to 1 month before the interest rate adjustment. One year after includes the month of the adjustment up to 11 months after. Two years after includes 12 months after the adjustment up to 23 months after. Column 1 examines the impact of mortgage interest rate resets on the scheduled monthly mortgage payment in dollars, while column 2 conducts similar analysis with scheduled monthly mortgage payments scaled by the initial monthly mortgage payment as the dependent variable. Column 3 examines the impact of mortgage interest rate resets on the monthly net dollar amount of new car purchases (financed with debt), while column 4 conducts similar analysis with net new car purchases scaled by the initial monthly mortgage payment as the dependent variable. Column 5 examines the relationship between the timing of mortgage interest rate resets and a dummy variable that takes the value of 1 in a month in which the borrower buys a new car and is 0 otherwise (reported in percentage terms). Column 6 examines the impact of mortgage interest rate resets on the voluntary monthly dollar repayment of mortgage debt, while column 7 conducts similar analysis with the voluntary repayment of mortgage debt scaled by the initial mortgage payment as the dependent variable. Other controls include a variety of borrower, mortgage, and regional characteristics including borrower FICO credit score, LTV ratio, zip-code-level house price controls, and fixed effects as discussed in the text. Standard errors (in parentheses) are clustered at the borrower and at the month level.

5.2 Table 2: Main borrower-level effects

Read:

- Monthly mortgage payments fall by **$943** in the first year after reset and **$938** in the second year.

- Scaled by the initial payment, the decline is about **52.8–52.9%**.
- New car spending rises by **$77** per month in the first year and **$108** per month in the second year.
- New car spending scaled by the initial payment rises by **4.3%** in the first year and **7.2%** in the second year.
- The car-purchase probability rises by **0.345 percentage points** in the first year and **0.478 percentage points** in the second year.
- Voluntary mortgage repayment rises by **$68** per month in the first year and **$75** per month in the second year.

Economic message. Households spend a meaningful share of the liquidity shock, but they do not spend all of it. They also use the shock to repair balance sheets by paying down mortgage debt.

5.3 Figure 1: Dynamic borrower responses around non-agency ARM resets

Read:

- Panel A shows mortgage payments in dollars. Payments are flat before reset and then fall sharply by close to $1,000.
- Panel B shows the same decline scaled by initial mortgage payment, close to one-half of the initial payment.
- Panel C shows new car spending. The increase is small before reset and then grows after reset, reaching large positive values several quarters later.
- Panel D shows voluntary mortgage repayment. Repayment rises after reset without a meaningful pre-trend.

Economic message. The timing is consistent with the mechanism: a mechanical payment reduction arrives first, and then households adjust consumption and deleveraging.

5.4 Table 3: Heterogeneous effects across households

Read:

- High-income borrowers receive similar payment reductions but have smaller car-spending responses than lower-income borrowers.
- High-LTV borrowers consume more out of the payment reduction and deleverage less, especially in the second year.
- High-FICO borrowers tend to deleverage more and have somewhat different consumption responses than lower-FICO borrowers.

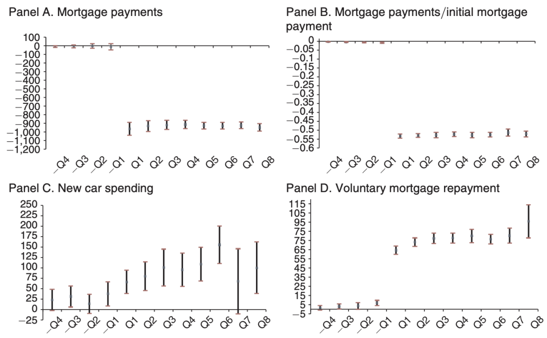

FIGURE 1. MONTHLY MORTGAGE PAYMENTS, NEW CAR SPENDING, AND VOLUNTARY DEBT REPAYMENT

Notes: This figure shows the OLS estimates for a set of quarterly time fixed effects along with 95 percent confidence intervals that allow us to assess the change in the relevant outcomes of borrowers with non-agency five-year ARMs during the four quarters preceding the interest rate reset as well as during each of the eight quarters following the rate reset. The dependent variables are the monthly mortgage payment in dollars (panel A), the mortgage payment scaled by the initial mortgage payment (panel B), new car spending in dollars (panel C), and voluntary repayment of mortgage debt in dollars (panel D).

TABLE 3—HETEROGENEOUS EFFECTS

	X = High income			X = High LTV			X = High FICO		
	Mortgage payment (1)	New car spending (2)	Debt repayment (3)	Mortgage payment (4)	New car spending (5)	Debt repayment (6)	Mortgage payment (7)	New car spending (8)	Debt repayment (9)
One year before	−0.00554	0.0273	6.86e-05	0.00132	−0.00850	0.000969	−0.00547	−0.00532	0.000264
	(0.000860)	(0.00879)	(0.00172)	(0.00147)	(0.0108)	(0.00179)	(0.000941)	(0.00853)	(0.00138)
One year after	−0.543	0.0706	0.0369	−0.520	0.0303	0.0417	−0.548	0.00312	0.0247
	(0.00580)	(0.0126)	(0.00313)	(0.00549)	(0.0145)	(0.00282)	(0.00562)	(0.0139)	(0.00268)
Two years after	−0.545	0.137	0.0435	−0.525	0.00982	0.0438	−0.547	0.0255	0.0326
	(0.00650)	(0.0219)	(0.00492)	(0.00768)	(0.0254)	(0.00601)	(0.00633)	(0.0288)	(0.00525)
(One year before) × X	0.00358	−0.0405	0.00165	−0.00958	0.0339	0.000368	0.00324	0.0178	0.00125
	(0.00175)	(0.00958)	(0.00217)	(0.00198)	(0.0140)	(0.00208)	(0.00166)	(0.0113)	(0.00172)
(One year after) × X	0.0303	−0.0529	0.00967	−0.0377	0.0485	−0.00814	0.0317	0.0592	0.0263
	(0.00314)	(0.0134)	(0.00394)	(0.00408)	(0.0214)	(0.00394)	(0.00324)	(0.0172)	(0.00327)
(Two years after) × X	0.0307	−0.124	0.00183	−0.0398	0.111	−0.000578	0.0290	0.0648	0.0176
	(0.00430)	(0.0252)	(0.00637)	(0.00491)	(0.0325)	(0.00681)	(0.00424)	(0.0302)	(0.00581)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,213,514	2,214,060	2,030,119	1,838,721	1,838,982	1,666,594	2,212,528	2,213,074	2,029,232
R^2	0.737	0.001	0.004	0.683	0.001	0.003	0.737	0.001	0.004

Notes: This table reports the coefficient estimates of OLS regressions relating the monthly mortgage payment, net new car spending, and voluntary mortgage principal repayments, all scaled by the monthly initial mortgage payment, to the timing relative to the first interest rate reset occurring five years after the loan's origination in a sample of borrowers with non-agency five-year ARMs. The main independent variables are dummies identifying different time periods before and after the reset date as well as their interaction terms with variables capturing borrower heterogeneity. One year before identifies the 12 months before up to 1 month before the interest rate adjustment. One year after includes the month of the adjustment up to 11 months after. Two years after includes 12 months after the adjustment up to 23 months after. Specifications in columns 1 to 3 include an additional independent variable High income (unreported for brevity) as well as its interaction with time dummies (reported in the table). High income is a dummy equal to 1 if the household's income, before the adjustment, is greater than the median income in our sample, and is 0 otherwise. Specifications in columns 4 to 6 include additional independent variable High LTV (unreported for brevity) as well as its interaction with time dummies (reported in the table). High LTV is a dummy equal to 1 if the loan's loan-to-value ratio one year before the adjustment is greater than 120 percent, and is 0 otherwise. Specifications in columns 7 to 9 include additional independent variable High FICO (unreported for brevity) as well as its interaction with time dummies (reported in the table). High FICO is a dummy equal to 1 if the FICO one year before the adjustment is greater than 660, and is 0 otherwise. Other controls include a variety of borrower, mortgage, and regional characteristics, as well as fixed effects in Table 2 as well as the aforementioned stand-alone dummies capturing the borrower heterogeneity. Standard errors in parentheses are clustered at the borrower and at the month level.

- The main heterogeneity patterns line up with the idea that low income and low housing wealth raise the marginal propensity to consume.

Economic message. Monetary-policy pass-through is not uniform. The same dollar of debt-service relief has larger consumption effects for financially weaker households.

5.5 Table 4 and Figure 2: External validity with conforming ARMs

TABLE 4—EXTERNAL VALIDITY: IMPACT OF RATE REDUCTIONS AMONG BORROWERS WITH CONFORMING ARMs

Panel A. Main effects

	Mortgage payment (1)	Mortgage payment/initial payment (2)	New car spending (3)	New car spending/initial payment (4)	Probability of new car spending (5)	Voluntary debt repayment (6)	Debt repayment/initial payment (7)
One year before	−6.37	−0.000	−0.44	0.000	0.007	1.07	0.001
	(6.28)	(0.002)	(8.14)	(0.008)	(0.028)	(0.39)	(0.000)
One year after	−276.57	−0.227	27.67	0.028	0.127	17.91	0.017
	(5.00)	(0.002)	(6.98)	(0.007)	(0.024)	(0.73)	(0.001)
Two years after	−279.87	−0.230	35.61	0.042	0.192	15.15	0.014
	(5.59)	(0.002)	(12.16)	(0.008)	(0.034)	(0.85)	(0.001)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,636,647	2,636,647	2,636,647	2,636,647	2,636,647	2,636,647	2,636,647
R^2	0.375	0.787	0.001	0.001	0.001	0.012	0.017

Panel B. Heterogeneous effects

	X = High income			X = High LTV			X = High FICO		
	Mortgage payment (1)	New car spending (2)	Debt repayment (3)	Mortgage payment (4)	New car spending (5)	Debt repayment (6)	Mortgage payment (7)	New car spending (8)	Debt repayment (9)
One year before	−0.000	−0.001	0.001	0.002	−0.005	0.001	−0.000	−0.001	0.000
	(0.002)	(0.009)	(0.001)	(0.000)	(0.010)	(0.000)	(0.002)	(0.011)	(0.000)
One year after	−0.223	0.034	0.017	−0.228	0.027	0.018	−0.222	0.032	0.006
	(0.002)	(0.009)	(0.001)	(0.002)	(0.010)	(0.001)	(0.001)	(0.010)	(0.001)
Two years after	−0.226	0.055	0.013	−0.233	0.026	0.015	−0.220	0.037	0.002
	(0.002)	(0.009)	(0.001)	(0.002)	(0.010)	(0.001)	(0.002)	(0.014)	(0.001)
(One year before) × X	0.000	0.003	0.001	−0.002	0.014	0.000	0.001	0.005	0.000
	(0.000)	(0.017)	(0.001)	(0.001)	(0.012)	(0.000)	(0.001)	(0.015)	(0.001)
(One year after) × X	−0.009	−0.010	0.000	0.000	0.002	−0.002	−0.008	0.002	0.015
	(0.001)	(0.014)	(0.001)	(0.000)	(0.012)	(0.001)	(0.001)	(0.013)	(0.002)
(Two years after) × X	−0.008	−0.025	0.002	0.004	0.037	−0.002	−0.015	0.021	0.016
	(0.001)	(0.015)	(0.001)	(0.001)	(0.012)	(0.002)	(0.001)	(0.025)	(0.002)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,636,647	2,636,647	2,636,647	2,636,647	2,636,647	2,636,647	2,636,647	2,636,647	2,636,647
R^2	0.787	0.001	0.017	0.788	0.001	0.012	0.788	0.001	0.011

Notes: Panel A of this table presents estimates from OLS regressions similar to ones displayed in Table 2 but estimated in a sample of borrowers with conforming five-year ARMs. The main independent variables are dummies identifying different time periods before and after the reset date. One year before identifies the 12 months before up to 1 month before the interest rate adjustment. One year after includes the month of the adjustment up to 11 months after. Two years after includes 12 months after the adjustment up to 23 months after. Panel B reports coefficient estimates of OLS regressions similar to ones displayed in Table 3 but estimated in a sample of borrowers with conforming five-year ARMs. High income, High LTV, and High FICO are dummy variables that take values of 1 for observations corresponding to borrowers with relatively high income, high loan-to-value ratios on their loans, and high FICO credit scores, and are 0 otherwise. Other controls include a variety of borrower, mortgage, and regional characteristics, and fixed effects similar to those in Table 2 and 3, respectively. Standard errors (in parentheses) are clustered at the borrower and at the month level.

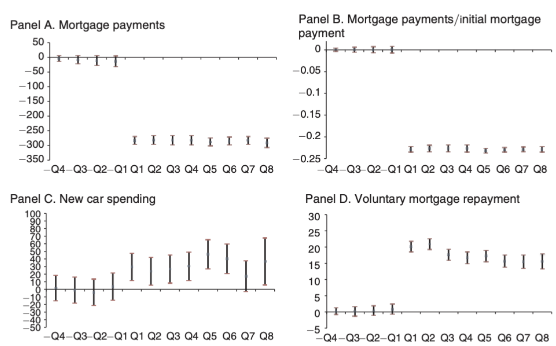

FIGURE 2. EXTERNAL VALIDITY: THE EFFECTS AMONG BORROWERS WITH CONFORMING (*Agency*) ARMs

Notes: This figure shows OLS estimates for a set of quarterly time fixed effects along with 95 percent confidence intervals that allows us to assess the change in the relevant outcomes of borrowers with conforming five-year ARMs during the four quarters preceding the first interest rate reset as well as during each of the eight quarters following the rate reset. The dependent variables are the monthly mortgage payment in dollars (panel A), the mortgage payment scaled by the initial mortgage payment (panel B), new car spending in dollars (panel C), and voluntary repayment of mortgage debt in dollars (panel D).

Read:

- In the conforming sample, monthly payments fall by about **$277** in the first year and **$280** in the second year after reset.
- This is about **22.7–23.0%** of the initial monthly payment.
- New car spending rises by about **$28** in the first year and **$36** in the second year.
- Voluntary mortgage repayment rises by about **$18** in the first year and **$15** in the second year.
- The timing in Figure 2 mirrors Figure 1: payment declines occur at reset, followed by positive car-spending and debt-repayment responses.

Economic message. The mechanism is not unique to large non-agency mortgages. It also appears among conforming borrowers, with magnitudes scaled down because their payment reductions are smaller.

5.6 Table 5: Difference-in-differences estimates

Read:

- For non-agency borrowers, five-year ARMs experience a differential payment decline of **$922** in the first quarter after reset relative to ten-year ARMs.
- Non-agency new car spending rises by roughly **$147–$187** per month in the first four quarters after reset in the DiD specification.
- Non-agency voluntary debt repayment rises by roughly **$66–$75** per month in the first several quarters after reset.
- For conforming borrowers, five-year ARMs experience a differential payment decline of about **$264–$277** in the year after reset relative to seven-year ARMs.
- Conforming borrowers increase car spending by about **$44–$74** per month and voluntary repayment by about **$14–$17** in the quarters after reset.

Economic message. The results are not driven by generic mortgage-age patterns. They appear specifically for the loans that mechanically reset.

5.7 Default, additional consumption measures, and other robustness checks

Read:

- The paper finds that lower mortgage payments reduce mortgage default rates.
- In the conforming sample, a reduction of monthly mortgage payments by roughly 20% reduces the likelihood of mortgage default over two years by about 40% relative to the mean delinquency rate.

- Additional evidence using store credit-card balances also points to increased consumption after reset.
- The paper examines repayments of home-equity loans and home-equity lines of credit as additional deleveraging margins.
- The paper discusses attrition, refinancing constraints, and the Home Affordable Refinancing Program as robustness and interpretation issues.

Economic message. The same balance-sheet relief that increases consumption also reduces financial distress. Payment reductions matter for both spending and default.

TABLE 5—DIFFERENCE-IN-DIFFERENCES ESTIMATES BASED ON THE ALTERNATIVE EMPIRICAL STRATEGY

	Borrowers with non-agency ARMs			Borrowers with conforming ARMs		
	Mortgage payment (1)	New car spending (2)	Voluntary debt repayment (3)	Mortgage payment (4)	New car spending (5)	Voluntary debt repayment (6)
Four quarters before	−10.75	40.11	8.44	−4.45	−6.82	0.28
	(3.158)	(22.13)	(3.781)	(5.08)	(21.12)	(0.73)
Three quarters before	−19.75	26.13	6.935	−7.83	8.79	0.84
	(4.999)	(31.34)	(5.476)	(6.01)	(23.54)	(0.86)
Two quarters before	−23.72	48.57	12.12	−8.05	24.91	1.11
	(5.701)	(33.82)	(6.623)	(6.89)	(20.68)	(0.92)
One quarter before	−20.74	99.45	10.61	−9.32	18.87	0.37
	(7.830)	(33.01)	(8.285)	(7.22)	(23.91)	(0.97)
One quarter after	−922.3	146.7	66.26	−276.52	43.73	16.73
	(43.13)	(44.50)	(10.89)	(8.28)	(23.19)	(1.15)
Two quarters after	−848.7	162.3	75.30	−272.23	52.43	17.31
	(33.92)	(46.14)	(13.43)	(8.67)	(22.07)	(1.25)
Three quarters after	−793.6	187.3	71.74	−268.93	73.70	14.41
	(33.68)	(40.96)	(15.72)	(8.71)	(22.24)	(1.22)
Four quarters after	−750.6	186.1	67.25	−264.37	62.69	14.44
	(35.07)	(60.76)	(17.86)	(8.94)	(22.37)	(1.37)
Two years after	−713.5	137.7	62.21	−254.48	71.60	12.96
	(36.82)	(79.58)	(20.15)	(6.26)	(22.26)	(1.33)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,565,771	3,622,209	3,424,364	3,398,806	3,398,806	3,398,806
R^2	0.977	0.030	0.243	0.53	0.001	0.012

Notes: This table presents OLS estimates from the difference-in-differences specifications that examine the evolution of the monthly dollar amount of scheduled mortgage payments, new car spending, and voluntary mortgage principal repayment for borrowers with five-year ARMs (the treatment group) relative to borrowers with ARMs facing a longer initial fixed-rate period (the control group). The main independent variables are dummies identifying different quarter periods before and after the loan's age of 60 months corresponding to the first reset date of five-year ARM contracts (not reported for brevity) and the interactions of these time dummies with a dummy variable for the five-year ARM contract type (reported in the table). Columns 1–3 show the estimates for a sample of borrowers with non-agency five-year ARMs relative to a control group of borrowers with non-agency ten-year ARMs. Columns 4–6 show the corresponding estimates for a sample of borrowers with conforming five-year ARMs relative to a control group of borrowers with conforming seven-year ARMs. Other controls include a variety of borrower, mortgage, and regional characteristics, fixed effects similar to those in Table 2, as well as the stand-alone dummy capturing the five-year ARM contract type. Standard errors (in parentheses) are clustered at the borrower and at the month level.

TABLE 6—REGIONAL EVIDENCE

Panel A. Summary statistics for high and low exposure zip codes

	High exposure zip codes		Low exposure zip codes	
	Mean	SD	Mean	SD
FICO	714.8	23.2	716.0	18.9
LTV ratio (percent)	64.5	7.29	68.1	7.00
Interest rate (percent)	6.64	0.57	6.68	0.48
Mortgage delinquency rate (percent)	2.81	3.09	2.23	1.83
Unemployment rate (percent)	6.04	1.55	5.91	1.47
Median income ($thousands)	58.4	14.13	52.7	14.38
Percentage of individuals with college degree	31.4	10.1	29.5	9.42
Percentage of households that are married couples with children	21.9	5.13	21.6	5.13
Consumer credit score	3.37	0.41	3.35	0.35
Quarterly house price growth (percent)	1.94	2.34	1.92	2.28
Percentage of GSE loans	69.0	9.61	70.2	10.7
ARM share	35.2	7.62	17.3	4.51

Panel B. Change in mortgage rates, mortgage delinquency, house price growth, auto sales growth, employment growth, and a zip code ARM share

	Mortgage interest rate		Mortgage delinquency growth rate	House price growth rate	Auto sales growth rate	Employment growth rate
	(1)	(2)	(3)	(4)	(5)	(6)
ARM share	−0.0198	−0.0176	−0.264	0.025	0.037	0.029
	(0.0005)	(0.0008)	(0.035)	(0.008)	(0.018)	(0.013)
Zip code controls	No	Yes	Yes	Yes	Yes	Yes
State fixed effects	No	Yes	Yes	Yes	Yes	Yes
R^2	0.56	0.75	0.69	0.49	0.28	0.11

Notes: Panel A presents average summary statistics of key variables in high exposure (above median ARM share) and low exposure (below median ARM share) zip codes. The sample consists of 1,000 zip codes as explained in Section VIIA. The key independent variable, ARM share, is the percentage of outstanding mortgages that are of ARM type in a zip code (as of mid-2006). Consumer credit score is the average consumer rating of borrowers in a zip code assigned by Equifax (see Section I for the discussion of data sources). Panel B reports OLS estimates from regressions where the dependent variable is the mean change in mortgage interest rates (column 1 and 2), the average quarterly mortgage delinquency growth rate (column 3), the house price growth rate (column 4), the auto sales growth rate (column 5), and the employment growth rate (column 6) in a zip code between the period of interest rate declines and the period preceding the rate declines. Zip code controls include average zip code mortgage LTV ratios, interest rates, house price controls, socio-economic variables capturing a profile of the zip code population, and the average credit score of households. State fixed effects are fixed effects for the state corresponding to the location of the zip code. The estimates are expressed in percentage terms; standard errors are in parentheses.

5.8 Table 6: Regional evidence

Read:

- High-exposure zip codes have average ARM shares of **35.2%**; low-exposure zip codes have average ARM shares of **17.3%**.
- High- and low-exposure zip codes are similar on many observables after matching: FICO scores, interest rates, unemployment, house-price growth, and GSE loan shares are close.
- A 10-percentage-point higher ARM share is associated with a roughly **17–20 basis point** larger decline in average mortgage rates.
- A 10-percentage-point higher ARM share is associated with a **0.264 percentage point** decline in quarterly delinquency growth.
- A 10-percentage-point higher ARM share is associated with a **0.25 percentage point** increase in quarterly house-price growth.

- A 10-percentage-point higher ARM share is associated with a **0.37 percentage point** increase in quarterly auto-sales growth and a **0.29 percentage point** increase in employment growth.

Economic message. The borrower-level channel aggregates up: areas with more exposure to automatic rate reductions experience stronger local spending, house-price, and employment outcomes.

6 Short summary

- The paper studies how mortgage-rate reductions affect households and local economies.
- Identification comes from automatic resets of five-year ARMs originated before the crisis.
- The main non-agency ARM reset lowers monthly payments by about $940.
- Borrowers respond by increasing debt-financed car purchases and by paying down mortgage principal faster.
- The consumption response is stronger among lower-income and higher-LTV borrowers.
- Similar but smaller effects appear in conforming agency ARMs.
- Difference-in-differences comparisons against longer-reset ARMs support the causal interpretation.
- Regional evidence shows that high-ARM-share areas experience stronger mortgage-rate pass-through and better local outcomes after interest rates fall.
- The paper's central lesson is that mortgage contract design shapes the household-balance-sheet channel of monetary policy.

7 Conclusion

7.1 Core conclusion

Di Maggio et al. show that lower mortgage rates have large effects when they actually pass through to household cash flows. Automatic ARM resets generate large, persistent reductions in required monthly payments. Households respond by consuming more, deleveraging more, and defaulting less.

7.2 Economic intuition

A lower required mortgage payment relaxes the household budget constraint every month. Some households spend the extra cash flow, especially on durable goods such as cars. Others use part of it to rebuild home equity and reduce debt. Financially weaker households have stronger consumption responses because their marginal utility of liquidity is higher.

7.3 Takeaway

Monetary policy is more powerful when falling rates reach borrowers directly. Mortgage-contract rigidity matters: fixed-rate mortgages require refinancing, while adjustable-rate mortgages can transmit rate cuts automatically. The paper therefore links monetary policy, mortgage design, household balance sheets, and local real activity in one empirical framework.